INCH-POUND
MIL-R-22684/1B
6 August 2001
SUPERSEDING
MIL-R-22684/1A
21 November 1968

MILITARY SPECIFICATION

RESISTORS, FIXED, FILM, INSULATED, STYLE RL07

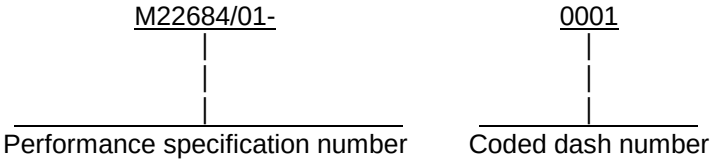
This specification is inactive for design after 21 November 1968. Use MIL-PRF-39017/1.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the associated requirements for insulated, film, fixed resistors of 2-percent and 5-percent resistance tolerance. These resistors are capable of full-load operation at an ambient temperature of 70°C and have a resistance-temperature characteristic of ±200 parts per million per degree Celsius (ppm/°C).

1.2 Part or Identifying Number (PIN). Resistors covered by this specification are identified by a PIN which consists of the basic number of this specification and a coded dash number taken from table I. The PIN is in the following form:



The coded dash number is derived in accordance with MIL-PRF-22684.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: DSCC-VAT, P.O. Box 3990 , Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the basic document or by letter.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-22684 - Resistors, Fixed, Film (Insulated), General Specification.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Service, Building 4D (DPM-DoDSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related association specification, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-22684.

3.2 Interface and physical dimensions. Resistors shall meet the interface and physical dimensions specified on figure 1, as applicable.

3.3 Power rating. The power rating shall be 1/4 watt based on full load operation at an ambient temperature of 70°C.

3.4 Voltage rating. The maximum continuous working voltage shall not exceed 250 volts.

3.5 Resistance values and resistance tolerances. The minimum and maximum standard resistance values and associated resistance tolerances shall be as listed in table I.

3.6 Terminal type. The terminal type available shall be in accordance with MIL-PRF-22684 and table I.

3.7 Dielectric withstanding voltage. Resistors shall be tested as specified in MIL-PRF-22684. The magnitude of test voltage shall be as follows:

Atmospheric pressure - 500 volts rms.
Barometric pressure - 325 volts rms.

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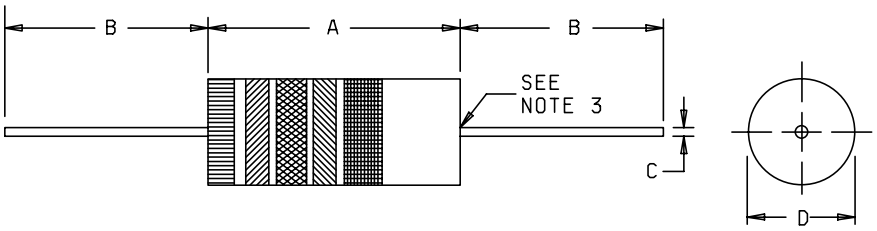

	DIMENSIONS			
	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX
A	.205	.281	5.56	7.14
B	1.375	1.625	34.92	41.28
C	.019	.027	0.48	0.68
D	.082	.098	2.08	2.49

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The end of the body shall be that point at which the diameter equals the nearest drill size larger than 250 percent of the nominal lead diameter. The leads shall be solderable to within .125 inch (3.18 mm) of the resistor body.

FIGURE 1. Configuration and dimensions.

3.8 Insulation resistance. Resistors shall be tested as specified in MIL-PRF-22684 except the insulation resistance shall be not less than 100 megohms.

3.9 Moisture resistance. Resistors shall be tested as specified in MIL-PRF-22684 except the change in resistance shall not exceed 1.5 percent.

3.10 Life. Resistors shall be tested as specified in MIL-PRF-22684 except the change in resistance shall not exceed 2.0 percent.

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Table I. PIN designation.

Type designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.	Type designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.
	Ohms					Ohms			
RL07S510G	51	G	S	0001	RL07S621G	620	G	S	0053
RL07S510J	51	J	S	0002	RL07S621J	620	J	S	0054
RL07S560G	56	G	S	0003	RL07S681G	680	G	S	0055
RL07S560J	56	J	S	0004	RL07S681J	680	J	S	0056
RL07S620G	62	G	S	0005	RL07S751G	750	G	S	0057
RL07S620J	62	J	S	0006	RL07S751J	750	J	S	0058
RL07S680G	68	G	S	0007	RL07S821G	820	G	S	0059
RL07S680J	68	J	S	0008	RL07S821J	820	J	S	0060
RL07S750G	75	G	S	0009	RL07S911G	910	G	S	0061
RL07S750J	75	J	S	0010	RL07S911J	910	J	S	0062
RL07S820G	82	G	S	0011	RL07S102G	1000	G	S	0063
RL07S820J	82	J	S	0012	RL07S102J	1000	J	S	0064
RL07S910G	91	G	S	0013	RL07S112G	1100	G	S	0065
RL07S910J	91	J	S	0014	RL07S112J	1100	J	S	0066
RL07S101G	100	G	S	0015	RL07S122G	1200	G	S	0067
RL07S101J	100	J	S	0016	RL07S122J	1200	J	S	0068
RL07S111G	110	G	S	0017	RL07S132G	1300	G	S	0069
RL07S111J	110	J	S	0018	RL07S132J	1300	J	S	0070
RL07S121G	120	G	S	0019	RL07S152G	1500	G	S	0071
RL07S121J	120	J	S	0020	RL07S152J	1500	J	S	0072
RL07S131G	130	G	S	0021	RL07S162G	1600	G	S	0073
RL07S131J	130	J	S	0022	RL07S162J	1600	J	S	0074
RL07S151G	150	G	S	0023	RL07S182G	1800	G	S	0075
RL07S151J	150	J	S	0024	RL07S182J	1800	J	S	0076
RL07S161G	160	G	S	0025	RL07S202G	2000	G	S	0077
RL07S161J	160	J	S	0026	RL07S202J	2000	J	S	0078
RL07S181G	180	G	S	0027	RL07S222G	2200	G	S	0079
RL07S181J	180	J	S	0028	RL07S222J	2200	J	S	0080
RL07S201G	200	G	S	0029	RL07S242G	2400	G	S	0081
RL07S201J	200	J	S	0030	RL07S242J	2400	J	S	0082
RL07S221G	220	G	S	0031	RL07S272G	2700	G	S	0083
RL07S221J	220	J	S	0032	RL07S272J	2700	J	S	0084
RL07S241G	240	G	S	0033	RL07S302G	3000	G	S	0085
RL07S241J	240	J	S	0034	RL07S302J	3000	J	S	0086
RL07S271G	270	G	S	0035	RL07S332G	3300	G	S	0087
RL07S271J	270	J	S	0036	RL07S332J	3300	J	S	0088
RL07S301G	300	G	S	0037	RL07S362G	3600	G	S	0089
RL07S301J	300	J	S	0038	RL07S362J	3600	J	S	0090
RL07S331G	330	G	S	0039	RL07S392G	3900	G	S	0091
RL07S331J	330	J	S	1040	RL07S392J	3900	J	S	0092
RL07S361G	360	G	S	0041	RL07S432G	4300	G	S	0093
RL07S361J	360	J	S	0042	RL07S432J	4300	J	S	0094
RL07S391G	390	G	S	0043	RL07S472G	4700	G	S	0095
RL07S391J	390	J	S	0044	RL07S472J	4700	J	S	0096
RL07S431G	430	G	S	0045	RL07S512G	5100	G	S	0097
RL07S431J	430	J	S	0046	RL07S512J	5100	J	S	0098
RL07S471G	470	G	S	0047	RL07S562G	5600	G	S	0099
RL07S471J	470	J	S	0048	RL07S562J	5600	J	S	0100
RL07S511G	510	G	S	0049	RL07S622G	6200	G	S	0101
RL07S511J	510	J	S	0050	RL07S622J	6200	J	S	0102
RL07S561G	560	G	S	0051	RL07S682G	6800	G	S	0103
RL07S561J	560	J	S	0052	RL07S682J	6800	J	S	0104

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Table I. PIN designation - continued.

Type designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.	Type designation	Nominal total resistance value	Resistance tolerance (percent)	Terminal	Dash No.
	Ohms					Ohms			
RL07S752G	7,500	G	S	0105	RL07S363G	36,000	G	S	0137
RL07S752J	7,500	J	S	0106	RL07S363J	36,000	J	S	0138
RL07S822G	8,200	G	S	0107	RL07S393G	39,000	G	S	0139
RL07S822J	8,200	J	S	0108	RL07S393J	39,000	J	S	0140
RL07S912G	9,100	G	S	0109	RL07S433G	43,000	G	S	0141
RL07S912J	9,100	J	S	0110	RL07S433J	43,000	J	S	0142
RL07S103G	10,000	G	S	0111	RL07S473G	47,000	G	S	0143
RL07S103J	10,000	J	S	0112	RL07S473J	47,000	J	S	0144
RL07S113G	11,000	G	S	0113	RL07S513G	51,000	G	S	0145
RL07S113J	11,000	J	S	0114	RL07S513J	51,000	J	S	0146
RL07S123G	12,000	G	S	0115	RL07S563G	56,000	G	S	0147
RL07S123J	12,000	J	S	0116	RL07S563J	56,000	J	S	0148
RL07S133G	13,000	G	S	0117	RL07S623G	62,000	G	S	0149
RL07S133J	13,000	J	S	0118	RL07S623J	62,000	J	S	0150
RL07S153G	15,000	G	S	0119	RL07S683G	68,000	G	S	0151
RL07S153J	15,000	J	S	0120	RL07S683J	68,000	J	S	0152
RL07S163G	16,000	G	S	0121	RL07S753G	75,000	G	S	0153
RL07S163J	16,000	J	S	0122	RL07S753J	75,000	J	S	0154
RL07S183G	18,000	G	S	0123	RL07S823G	82,000	G	S	0155
RL07S183J	18,000	J	S	0124	RL07S823J	82,000	J	S	0156
RL07S203G	20,000	G	S	0125	RL07S913G	91,000	G	S	0157
RL07S203J	20,000	J	S	0126	RL07S913J	91,000	J	S	0158
RL07S223G	22,000	G	S	0127	RL07S104G	100k	G	S	0159
RL07S223J	22,000	J	S	0128	RL07S104J	100k	J	S	0160
RL07S243G	24,000	G	S	0129	RL07S114G	110k	G	S	0161
RL07S243J	24,000	J	S	0130	RL07S114J	110k	J	S	0162
RL07S273G	27,000	G	S	0131	RL07S124G	120k	G	S	0163
RL07S273J	27,000	J	S	0132	RL07S124J	120k	J	S	0164
RL07S303G	30,000	G	S	0133	RL07S134G	130k	G	S	0165
RL07S303J	30,000	J	S	0134	RL07S134J	130k	J	S	0166
RL07S333G	33,000	G	S	0135	RL07S154G	150k	G	S	0167
RL07S333J	33,000	J	S	0136	RL07S154J	150k	J	S	0168

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with Group A inspection and Group B inspection of MIL-PRF-22684 and as specified herein.

5. PACKAGING

5.1 Packaging. For acquisition purposes the packaging requirements shall be as specified in the contract or purchase order. When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes The notes specified in MIL-PRF-22684 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, date of this specification, and complete PIN (see 1.2).

b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2).

c. Packaging requirements.

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Army - AR, MI
Air Force - 19

Preparing activity:
DLA - CC

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete block1s 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** **MIL-R-22684/1B**	**2. DOCUMENT DATE** (YYMMDD) 2001 August 6

3. DOCUMENT TITLE
RESISTORS, FIXED, FILM, INSULATED, STYLE RL07

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Incl Area Code) (1) Commercial (2) DSN (If applicable)	7. DATE SUBMITTED (YYYYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Supply Center, Columbus ATTN: DSCC – VAT	b. TELEPHONE (Include Area Code) (1) Commercial (614) 692-0553 (2) DSN 850-0553
c. ADDRESS (Include Zip Code) P.O. 3990 Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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